SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CONCEPT MAPPING

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	**Assessment:**	Assessment Tool 3 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (5 Questions × 100 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	**Syed Arshad Khadri**	**Reg. No.:**	**192525255**
Section / Batch:	**B.Tech AI-ML**	**Date:**	**06-10-2026**

Question Type	Focus Area	Questions	Marks
Concept Mapping	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling - star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2, Q3, Q4, Q5	100
		GRAND TOTAL:	**100 Marks**

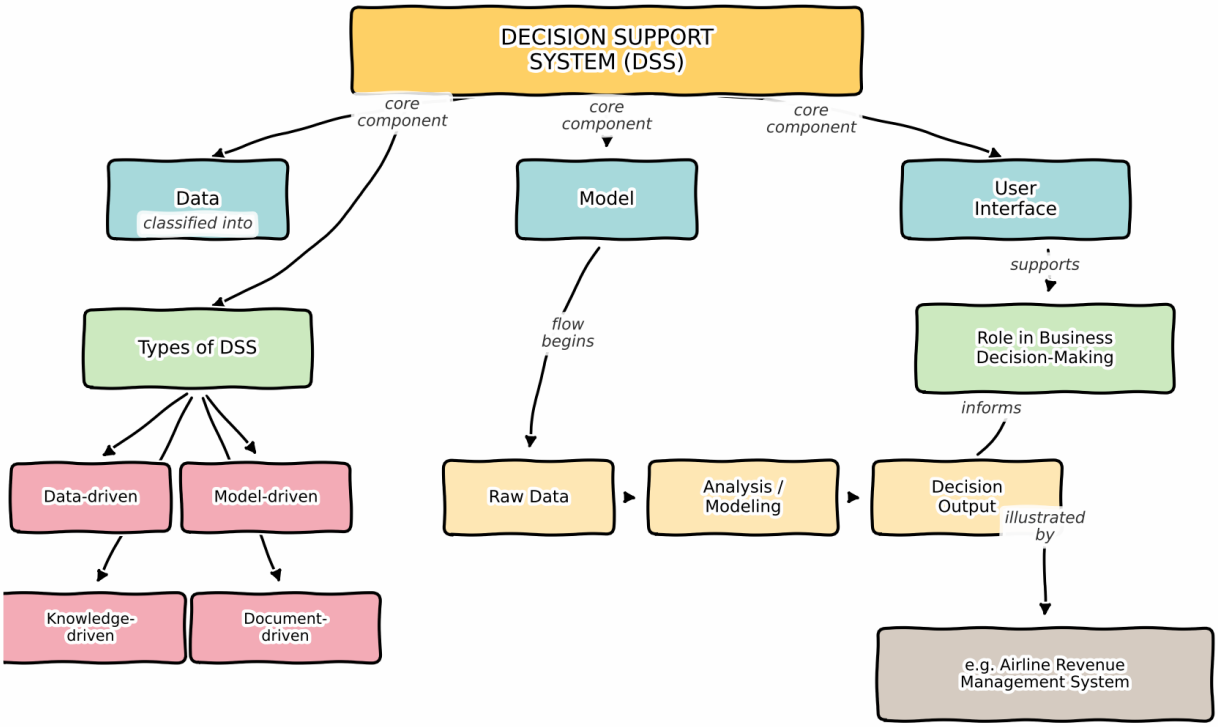
SECTION A – Technical Presentation

Concept Map 1: Decision Support Systems (DSS)

Task Description: *Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.*

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage

Concept Map (hand-drawn style):

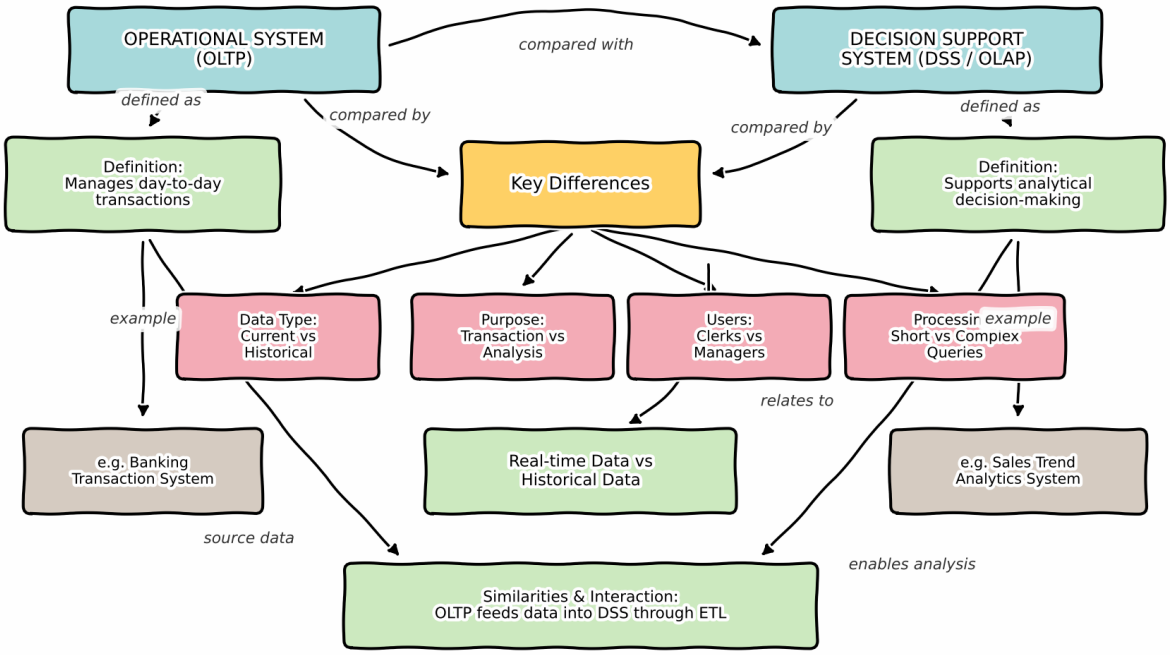


Concept Map 2: Operational Systems vs DSS

Task Description: *Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).*

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map (hand-drawn style):

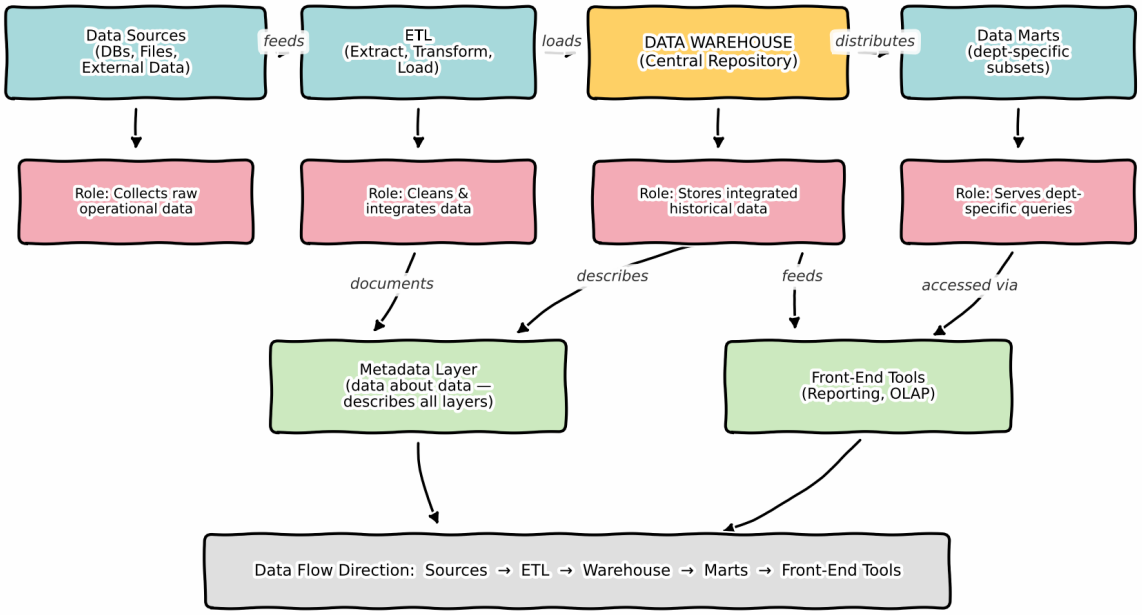


Concept Map 3: Architecture of Data Warehouse

Task Description: *Create a hierarchical concept map illustrating the architecture of a Data Warehouse.*

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map (hand-drawn style):

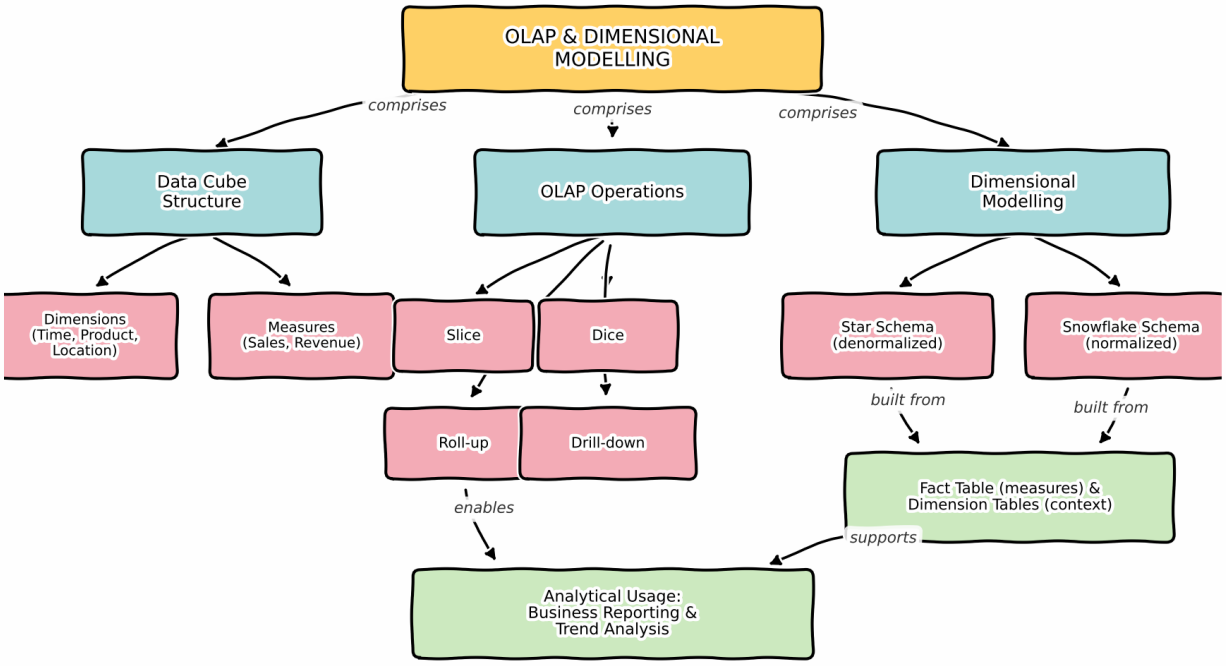


Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

Task Description: *Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.*

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map (hand-drawn style):

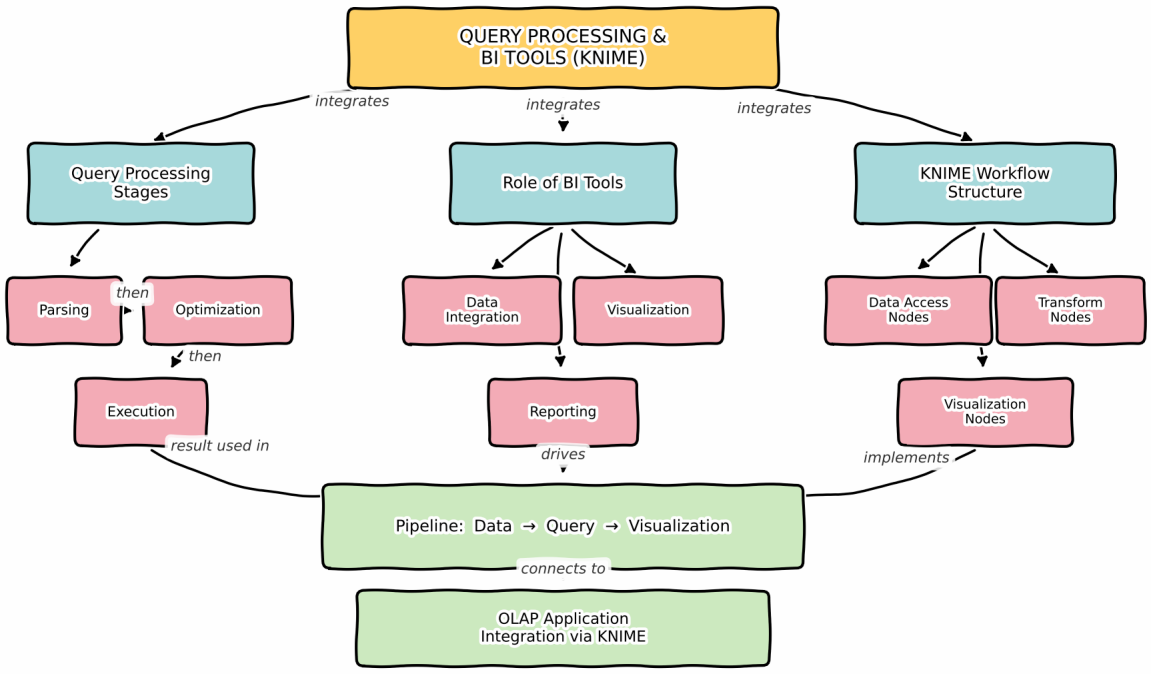


Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description: *Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.*

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

Concept Map (hand-drawn style):



Code of Conduct

I **Syed Arshad Khadri (192525255)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria (Wt.)	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification (25%)	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages (25%)	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization (20%)	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature (20%)	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity (10%)	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Marks Table

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%) Total (Each – 50)
Q1					
Q2					
Q3					
Q4					
Q5					
Overall Total (100)					